Fertility and Housing Tenure Choice

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Housing and Fertility

Housing Market | Real Estate 4 minute read

US birth rate hits record low, housing costs weigh on family planning

The 2008 financial crisis pushed both housing and fertility

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News Article

US Birth Rate Warning Issued Over House Prices

Want to Boost Birthrates? Build More Homes

Policymakers looking to reduce childcare costs should work to expand housing supply.

POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.

Renting a home 'leads to lower birth rates'

Housing, Fertility, and Tenure Choice

Children increase how much physical space a household needs. Access to that space depends on tenure and wealth:

- Rental housing can limit the size of the home a family can occupy.
- Ownership expands the housing menu but requires liquid wealth for a down payment.
- Older owners may retain family-sized homes after their children leave.
- Young families and older households compete for housing: its allocation across generations matters for fertility.

How much does access to family-sized housing affect fertility?
Which housing constraints matter, and for whom?

This Paper

Need a framework where fertility, tenure, housing size, saving, and population are jointly determined.

Model: overlapping generations with fertility, tenure, and saving

- Income heterogeneity, earnings risk, and bequests.
- Rental size limits, down payments, and housing transaction costs.
- Childbirth raises housing needs and changes future housing demand.

Empirics: housing around births, birth timing, and wealth by age

Policy: compare housing reforms along a demographic transition

- An externally specified decline in fertility preferences starts the transition.
- Introduce the reform along the path, from the same inherited state.

Quantitative Model

Environment

- Overlapping generations of households in one housing market.
- Households live for a finite number of periods, work, and then retire.
- Earnings differ permanently across households and are subject to persistent idiosyncratic risk.
- Households choose whether to have a child during their fertile years.
- Children depend on their parents until they mature.
- Households choose tenure, housing size, consumption, and saving throughout life.

Preferences

Households value consumption c, housing services s, and children at home m:

$$u_t(c, s; m) = e(m)^{\sigma-1} \frac{\left[c^{\alpha_0}\left(s - \bar{h}(m)\right)^{1-\alpha_0}\right]^{1-\sigma}}{1-\sigma} + \psi_t m.$$

Family size raises household needs through the equivalence scale $e(m)$ and the space requirement $\bar{h}(m)$.

- α_0 is the consumption share; σ governs curvature.
- ψ_t is the preference for children at date t.
- The equivalence scale $e(m)$ (Scholz et al., 2006) is increasing and concave.
- Its shape is fixed externally and normalized to a childless couple.

Family Space

The first child creates a discrete increase in required housing services:

$$\bar{h}(m) = h_1 \mathbf{1}\{m > 0\} + h_m m.$$

h_1 is the first-child requirement; h_m is the additional requirement per child.

Housing services depend on tenure:

$$s = \begin{cases} h^R, & \text{renter}, \\ \chi h', & \text{owner}, \end{cases} \qquad \chi > 1.$$

Owners obtain a service premium χ from a home of physical size h'.

Child needs disappear as children mature; lifetime births remain in the state.

Earnings and Bequests

Earnings $y(a, z)$ combine an age profile and persistent income risk (De Nardi, 2004), with permanent differences across households. Households receive retirement income after working life.

At death, households value estate wealth w^e:

$$\mathcal{B}(w^e) = \theta_0 \frac{(\theta_1 + \max\{w^e, 0\})^{1-\sigma}}{1-\sigma}.$$

θ_0 governs the bequest motive; θ_1 governs its sensitivity to wealth.

- Estate wealth includes liquid assets and housing wealth.
- Bequests affect saving and old-age housing retention.
- Entrant wealth is externally distributed; the model is not fully dynastic.

Housing and Tenure

Rental housing is continuous up to a size limit; owned housing comes in discrete sizes:

$$h^R \in [0, \bar{h}^R], \qquad h' \in \{0\} \cup \mathcal{H}^O, \qquad h' h^R = 0.$$

A household either rents or owns, and the largest owned homes exceed the rental limit.

- Buying requires a down payment of $(1 - \phi) P_t h'$.
- P_t is the purchase price per room; ϕ is the financed share.
- Selling incurs a proportional transaction cost ψ^s.
- Borrowing is secured by housing; the debt limit tightens with age.

Budget Constraints

Changing inherited owner size h to h' changes liquid resources by

$$\mathcal{T}_t(h, h') = \mathbf{1}\{h' \neq h\} \left[(1 - \psi^s) P_t h - P_t h'\right].$$

The household pays for its new home and receives net proceeds from the old one.

For renters and owners, respectively,

$$c + r_t h^R + b' = R_b b + y(a, z) + \mathcal{T}_t(h, 0) + T_t,$$

$$c + (\delta_H + \tau_t^p) P_t h' + b' = R_b b + y(a, z) + \mathcal{T}_t(h, h') + T_t.$$

R_b is the gross financial return, r_t rent per room, δ_H depreciation, τ_t^p the property-tax rate, and T_t a transfer per household head.

Households carry liquid wealth b into the period and choose saving b'.

Sequential Fertility and Child Aging

- During fertile ages, a household below terminal parity chooses whether to attempt the next birth.
- An attempt succeeds with age-specific probability π_a.
- A successful birth:

  $$n' = n + 1, \qquad m' = m + 1,$$

  and the first successful birth pays the one-time utility cost F_1.
- Fertility taste shocks have scale κ_1 on the first-birth margin and κ_C on continuation margins.
- Each child at home matures independently; m falls as children leave, while parity n remains a stock.

A birth changes the household's space needs immediately, but its demographic effects propagate across cohorts.

Housing Supply and Prices

Physical housing supply responds to the purchase price:

$$H_t^s = H_0 \left(\frac{P_t}{\bar{P}} \right)^{\eta}.$$

H_0 is supply at reference price $\bar{P}$; η is the supply elasticity.

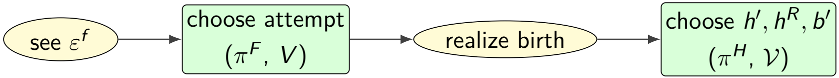
Rent and purchase prices satisfy the asset-pricing condition:

$$r_t + P_{t+1} = (R_b + \delta_H + \tau_t^p) P_t.$$

The current rent depends on both the carrying cost and the anticipated resale price.

Housing is supplied separately at each date; there is no construction stock in this specification.

The Household's Problem: Sequential Choices



Households choose fertility before learning whether conception succeeds.

1. Observe fertility tastes and choose whether to attempt a birth.
2. Learn whether the attempt succeeds.
3. Choose tenure and housing size, given the realized family size.
4. Choose consumption and saving, anticipating future income and family needs.

The Household's Problem: Housing and Saving

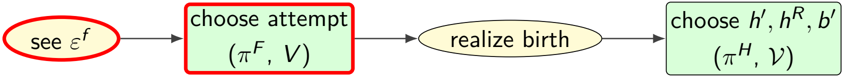
see ε^f $\rightarrow$ choose attempt (π^F, V) $\rightarrow$ realize birth $\rightarrow$ choose h', h^R, b' $(\pi^H, \mathcal{V})$

For a household of age a, given its wealth, earnings, housing, and realized family size:

$$\mathcal{V}_t = \mathbb{E}_{\varepsilon_t^H} \max_{h', h^R, b'} \left\{ u_t(c, s; m) + \varepsilon_t^H(h') + \beta s_a \mathbb{E}_t[V_{t+1}] + \beta(1 - s_a)\mathbb{E}_t[\mathcal{B}(w_{t+1}^e)] \right\}.$$

- $h' h^R = 0$: a renter sets $h' = 0$; an owner sets $h^R = 0$.
- Current purchases occur at P_t and future sales at future prices.
- The budget, down-payment condition, and age-tapered debt floor restrict (h', h^R, b').
- β discounts the future; s_a is survival; V_{t+1} is the next-period value.
- Housing-product taste shocks ε_t^H have scale κ_H.

The Household's Problem: Fertility Choice



Let $\mathcal{V}_t^+$ denote the value with an additional child and π_a conception success. The values of waiting and attempting are:

$$W_t^0 = \mathcal{V}_t, \qquad W_t^A = \pi_a\big[\mathcal{V}_t^+ - F_1\mathbf{1}\{n=0\}\big] + (1-\pi_a)\mathcal{V}_t.$$

The attempt probability is

$$\pi_t^F = \frac{\exp\{W_t^A/\kappa(n)\}}{\exp\{W_t^A/\kappa(n)\} + \exp\{W_t^0/\kappa(n)\}}, \qquad \kappa(n) = \begin{cases} \kappa_1, & n = 0, \\ \kappa_C, & n \geq 1. \end{cases}$$

Housing affects fertility through current child-space needs, tenure feasibility, and the continuation value of wealth and housing.

Population and Households

Births change future housing demand by changing the number and age of households.

- N_t: resident persons, tracked by age and sex.
- H_t: household heads, obtained from persons using age–sex headship rates.
- G_t: the distribution of household wealth, earnings, tenure, and children.

Births enter the population immediately. Survival, migration, and household formation determine when they add to housing demand.

▸ Population accounting

Transition Equilibrium

Given the inherited household and population state and the announced preference and policy paths, a perfect-foresight equilibrium is a sequence

$$\{V_t, \pi_t, G_t, N_t, H_t, P_t, r_t, T_t\}_{t\geq 0},$$

where V_t are household values and π_t are choice probabilities, satisfying:

1. Household optimality given the anticipated price and transfer paths.
2. Population and household evolution under births, survival, and migration.
3. Housing-market clearing: occupied physical rooms equal H_t^s.
4. Asset pricing and the specified government budget rule.

Aggregate paths are anticipated; individual outcomes remain uncertain.

Steady-State Equilibrium

A steady state is a constant allocation, population structure, and price:

$$(N^*, H^*, G^*, \pi^*, P^*, r^*, T^*).$$

Households optimize, persons and household distributions reproduce themselves, housing clears, and the fiscal rule holds.

- With no migration, births must replace deaths in a constant population.
- With migration, births and net inflows together offset deaths.
- A constant continuation value alone does not make a finite path stationary.

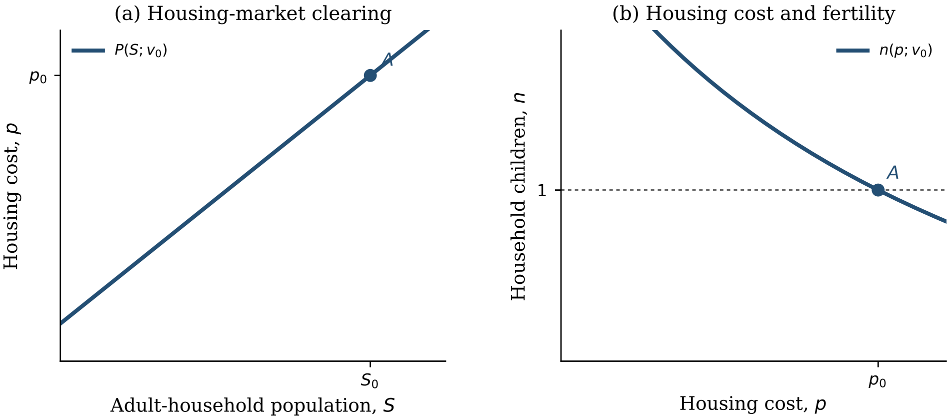
Housing and Fertility in Equilibrium

The next three diagrams illustrate the interaction between housing costs and population.

- **Left panel:** $P(S; v)$ is the housing cost that clears the market at adult-household population S, holding age composition fixed.
- **Right panel:** $n(p; v)$ is household fertility at housing cost p and fertility preference v.
- The illustration uses a closed population, with replacement equal to $n = 1$.

These are schematic schedules. The quantitative model tracks the full age distribution and distinguishes rents from asset prices.

Initial Steady State



At A, housing clears at (S_0, p_0) and fertility is at replacement.

The housing cost and population are constant in this schematic initial economy.

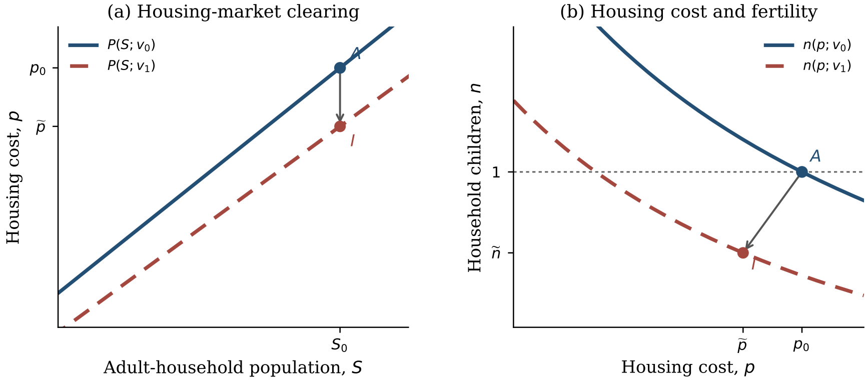
A Decline in Fertility Preferences

Begin from an approximate initial steady state. At the start of the transition, households learn the full future preference path.

- The preference for children declines exogenously over 2007–2023.
- The preference level remains fixed thereafter.
- Prices and household choices can respond as soon as the path is known.
- Liquid asset holdings, inherited homes, and cohort sizes are fixed at impact.

The exercise conditions on this decline; explaining its original cause is outside the question.

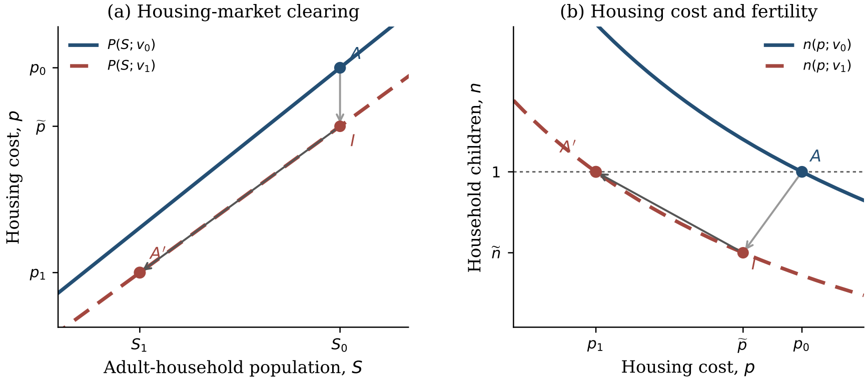
Impact



A fall from v_0 to v_1 shifts both schedules. With S_0 inherited, the schematic impact equilibrium moves from A to I: housing cost and fertility fall.

The diagram shows a one-time shift; the quantitative preference decline is spread over time.

Demographic Adjustment



Smaller entering cohorts can reduce housing demand. Lower housing costs partly offset the preference decline; A' marks a possible stationary outcome.

The arrow links the illustrated equilibria. Changing age composition can produce overshooting; it is not a computed transition or a convergence result.

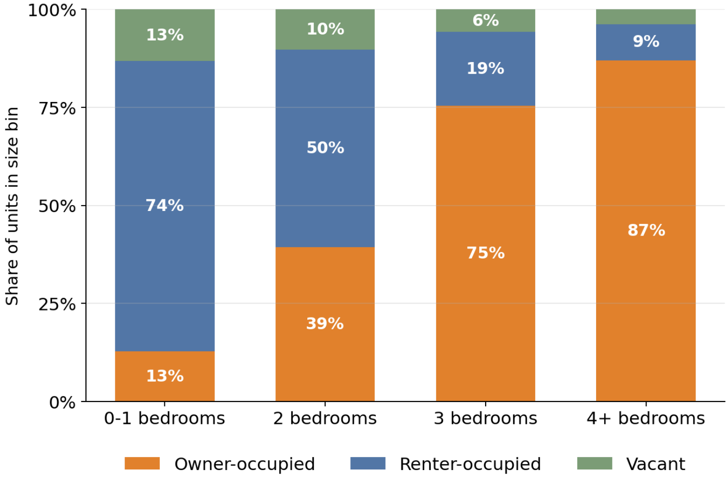
Empirics and Quantification

Data

- **AHS and ACS:** housing sizes, tenure, household composition, and age profiles.
- **PSID:** housing around childbirth, earnings, and wealth by age.
- **CPS and natality records:** completed fertility, childlessness, and birth timing.
- **Census population estimates:** age–sex population and demographic flows.

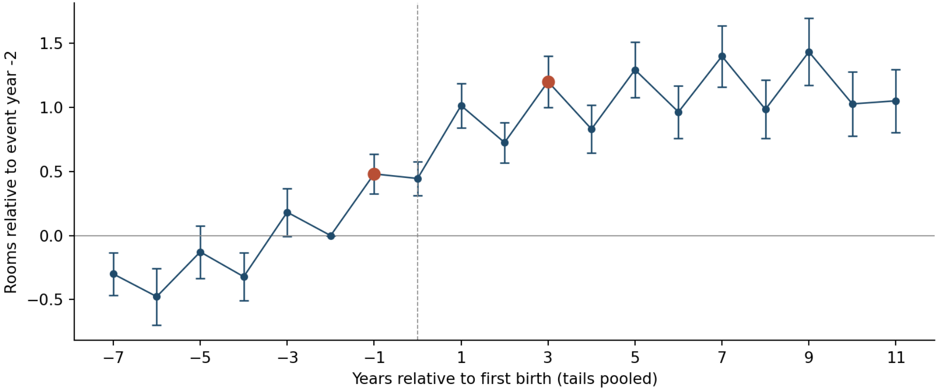
Cross-sectional housing patterns discipline access to space; longitudinal data describe housing adjustment around births.

Big Units Are Owned



Source: American Housing Survey, 2023.

Housing Size Around First Birth



Rooms increase around first birth, including before the birth. The four-year contrast from -1 to $+3$ is descriptive, not a causal effect.

Source: PSID; Sun–Abraham event study, household-aligned sample; pointwise 95% intervals.

Empirical Discipline

Different moments inform the fertility, housing, and saving decisions jointly.

Margin	Observations	Parameters informed
Fertility	Childlessness, completed fertility, first-birth timing	ψ, F_1, κ_1, κ_C
Housing	Rooms around births, family-size room gaps, ownership	h_1, h_m, χ
Saving and bequests	Wealth/earnings, bequest flows, old-age wealth dispersion	β, θ_0, θ_1
Housing supply	Housing quantities and price responses	H_0, η

A moment can inform several parameters; identification depends on their joint effects.

The Initial Economy and the Transition

- Use an approximate pre-2007 stationary economy for inherited household states.
- The chosen initial fertility benchmark is 2.1.
- Introduce the announced preference decline and compare dated model outcomes with the data.
- Treat the 2023 economy as a point along the transition.

Period birth rates, completed cohort fertility, and births per household head are distinct measurements.

Housing Policy

Policy Along the Transition

At date t_p, introduce a housing reform after the fertility-preference decline has begun.

Both paths start from the same state immediately before that announcement:

$$(G_{t_p}, N_{t_p}, H_{t_p})^{\text{reform}} = (G_{t_p}, N_{t_p}, H_{t_p})^{\text{baseline}}.$$

The inherited liquid assets, homes, and age distributions are therefore identical.

- Keep preferences, survival, migration, and the entry rule common.
- Preserve the housing-supply schedule except when it is the policy instrument.
- Solve a new anticipated price and allocation path after the announcement.

The Rebated Property Tax

Consider a permanent property-tax increase with contemporaneous revenue returned equally to household heads:

$$T_t \int dG_t = \tau_t^p P_t Q_t^H, \qquad Q_t^H = \int q_t(x)\, dG_t(x).$$

Here $q_t(x)$ is occupied physical housing, including rented rooms; τ_t^p is the tax rate per model period.

- **Capitalization:** changes in carrying costs affect asset prices.
- **Access:** prices and transfers change down-payment affordability.
- **Household choices:** tenure, housing size, saving, and fertility adjust.
- **Demography:** different births change future cohorts and housing demand.

Compare paths under the same rebate rule, with a common inherited state and supply schedule.

Conclusion

- Children raise demand for rooms; rental size and down payments govern access to that space.
- Tenure, saving, and fertility interact over the household lifecycle.
- Births change population and housing demand with long delays.
- Housing policy changes choices along an ongoing demographic transition.

The policy comparison is between paths starting from the same inherited economy.

Thank You

Appendix

The Person-Cohort Law

Let $N_t^{g,a}$ denote resident persons of sex g and single year of age a.

The annual person law is

$$N_{t+1} = A_{t+1} N_t + f_{t+1} B_{t+1} + M_{t+1}.$$

- A_{t+1} ages existing cohorts using destination-age survival.
- $f_{t+1} B_{t+1}$ allocates newborn survivors by sex at age zero.
- M_{t+1} is the age–sex net-migration flow after survival.
- Model-period births are assigned to calendar cohorts before the annual law is iterated.

Births and household heads remain distinct objects throughout the transition.

The Household-Head Law

Fixed age–sex headship rates $\rho^{g,a}$ map persons into the target head stock:

$$H_{t+1}^{g,a} = \rho^{g,a} N_{t+1}^{g,a} = H_{t+1}^{\mathrm{surv},g,a} + F_{t+1}^{g,a} - D_{t+1}^{g,a} + H_{t+1}^{M,g,a}.$$

The household operator supplies a raw within-age distribution $\widetilde{G}_{t+1,a}$. Its mass is rescaled to the person-law head target:

$$G_{t+1,a}(x) = H_{t+1,a} \frac{\widetilde{G}_{t+1,a}(x)}{\int \widetilde{G}_{t+1,a}(x')\, dx'}.$$

- F and D are the minimum one-way formation or dissolution flow needed to attain $\rho^{g,a}$.
- Wealth, tenure, income, parity, and child-state composition are preserved within age.
- This is an accounting closure, not an estimated household-formation hazard.

Exact Annual Person Accounting

Let $s^{g,a}$ be survival and f^{g} the birth sex share. For newborns,

$$N^{g,0}_{t+1} = s^{g,0}_{t+1} f^{g}_{t+1} B_{t+1} + M^{g,0}_{t+1}.$$

For interior ages $a \geq 1$,

$$N^{g,a}_{t+1} = s^{g,a}_{t+1} N^{g,a-1}_{t} + M^{g,a}_{t+1}.$$

The terminal age is top coded, so survivors from the penultimate and terminal cells both enter the last cell. Every step satisfies

$$\sum N_{t+1} = \sum N_t + B_{t+1} - \text{newborn deaths} - \text{existing deaths} + \sum M_{t+1}.$$

The Within-Age Household Bridge

Let $\widetilde{G}_{t+1,a}(x)$ be the raw household transition within age a, and let $H_{t+1,a}$ be the head mass implied by persons. Then

$$G_{t+1,a}(x) = \begin{cases} H_{t+1,a} \dfrac{\widetilde{G}_{t+1,a}(x)}{\int \widetilde{G}_{t+1,a}(x')\, dx'}, & \int \widetilde{G}_{t+1,a} > 0, \\ \\ H_{t+1,a} \Gamma_a(x), & \text{for an empty entrant age cell,} \end{cases}$$

where Γ_a is the externally specified entrant template. The operation exposes added and removed mass separately and moves no mass across ages.

A richer model would estimate economic states for newly formed, dissolved, and migrant heads rather than preserve the raw conditional composition.

Housing-Market Clearing

Let $q_t(x)$ denote the physical housing occupied after tenure choice. Market clearing requires

$$Q_t^H = \int q_t(x)\, dG_t(x) = H_0 \left(\frac{P_t}{\bar{P}} \right)^{\eta}.$$

For renters, $q_t = h_t^R$; for owners, $q_t = h_t'$. Choice probabilities are included when aggregating.

- The owner-service premium changes utility, not physical supply.
- Conditional renter policies enter demand only when renting is chosen.
- The same supply schedule applies to baseline and tax-reform paths.

Perfect-Foresight Solution

1. Guess the entire price path and compute rents from asset pricing.
2. Solve households backward using date-specific continuation values.
3. Advance household and population distributions from the inherited state.
4. Update prices until housing clears at every date.
5. Extend the horizon and check that the measured outcomes are stable.

A stationary terminal value supplies a boundary condition; it does not establish stationarity of the simulated endpoint.